time the measurement is repeated. Averaging several measurements will *always* improve the precision. In short, *precision is a measure of random noise.*

Now, imagine a measurement that is very precise, but has poor accuracy. This makes the histogram very slender, but not centered over the true value. Successive readings are close in value; however, they *all* have a large error. Poor accuracy results from **systematic errors.** These are errors that become repeated in exactly the same manner each time the measurement is conducted. Accuracy is usually dependent on how you *calibrate* the system. For example, in the ocean depth measurement, the parameter directly measured is elapsed time. This is converted into depth by a calibration procedure that relates *milliseconds* to *meters*. This may be as simple as multiplying by a fixed velocity, or as complicated as dozens of second order corrections. Averaging individual measurements does nothing to improve the accuracy. In short, *accuracy is a measure of calibration.*

In actual practice there are many ways that precision and accuracy can become intertwined. For example, imagine building an electronic amplifier from 1% resistors. This tolerance indicates that the value of each resistor will be within 1% of the stated value over a wide range of conditions, such as temperature, humidity, age, etc. This error in the resistance will produce a corresponding error in the gain of the amplifier. Is this error a problem of accuracy or precision?

The answer depends on how you take the measurements. For example, suppose you build *one* amplifier and test it several times over a few minutes. The error in gain remains constant with each test, and you conclude the problem is *accuracy*. In comparison, suppose you build *one thousand* of the amplifiers. The gain from device to device will fluctuate randomly, and the problem appears to be one of *precision*. Likewise, any one of these amplifiers will show gain fluctuations in response to temperature and other environmental changes. Again, the problem would be called *precision*.

When deciding which name to call the problem, ask yourself two questions. First: Will averaging successive readings provide a better measurement? If yes, call the error precision; if no, call it accuracy. Second: Will calibration correct the error? If yes, call it accuracy; if no, call it precision. This may require some thought, especially related to how the device will be calibrated, and how often it will be done.